

Tests & Quizzes

Question Pool: Quiz 02

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Question 1: Single Correct - 1.0 point

In the context of the Byzantine Generals Problem, which of the following situations does NOT represent a general acting maliciously or deceitfully?

- ☐ OA. A general sending different battle plans to different generals.
- ☐ OB. A general issuing misleading instructions to create confusion.
- ☒ OC. A general transmitting orders after a significant communication delay.
- ☐ OD. A general deliberately providing false commands while appearing cooperative.

Answer Key: C

Question 2: Single Correct - 1.0 point

Why can neither Proof of Work nor Proof of Stake guarantee consensus if a majority of influence becomes malicious?

- ☐ OA. Because cryptographic verification becomes unreliable beyond that point.
- ☒ OB. Because the group having majority of influence has the power to determine the accepted chain.
- ☐ OC. Because block validation becomes computationally infeasible.
- ☐ OD. Because peer-to-peer networking protocols collapse.

Answer Key: B

Question 3: Single Correct - 1.0 point

Consider the following sequence in a blockchain transaction lifecycle:

1. A transaction is created and broadcast to the network.
2. The transaction is validated and grouped with other transactions.
3. Network participants reach agreement on a proposed block, establishing a new authoritative state.
4. The agreed block is propagated and stored locally by all nodes.

At which step is the ledger logically updated?

- ☐ OA. Step 1.
- ☐ OB. Step 2.
- ☒ OC. Step 3.
- ☐ OD. Step 4.

Answer Key: C

Question 4: Single Correct - 1.0 point

A system has **multiple writers**, and all writers are **known and trusted entities**. Which design choice is most appropriate?

- ☐ OA. Public permissionless blockchain.
- ☐ OB. Private permissioned blockchain.
- ☐ OC. Public permissioned blockchain.
- ☒ OD. Do not use a blockchain.

Answer Key: D

Question 5: Single Correct - 1.0 point

You are validating transactions and constructing a block. Before you broadcast your block, you receive a valid block mined by another participant. That block has been broadcast but is not yet finalized by the network. What is the **most appropriate action** you should take?

- ☐ OA. Immediately broadcast your own block to compete for acceptance.
- ☒ OB. Discard your block and begin validating on top of the received block.
- ☐ OC. Pause your current work and resume validation only after the received block is finalized.
- ☐ OD. Continue mining your block until a fork is explicitly detected.

Answer Key: B

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